

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Dynamic equilibria, feedback & tipping points - Set 1

Mark scheme

AQA spec 3.6.1; Sustainability; textbook pp. 370-372

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark.</i> • Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs (1)	1
01.2	<i>1 mark for each point.</i> • Impact depends on total population x per capita impact (1) • With few people, natural processes (forest regrowth, fish recruitment, aerobic decomposition / dilution) recover faster than the resource is used or the waste added (1) • Larger populations exceed the rate of natural recovery: forest and soil do not regenerate, catches exceed MSY, water deoxygenates (1)	3
01.3	<i>1 mark for each point.</i> • Urban areas are supported by resources brought in from elsewhere; when one area is depleted, exploitation simply moves (timber, fish, minerals, energy) so consumers see no shortage (1) • Rural communities exploit local resources directly so the effects of over-exploitation become obvious quickly, encouraging cultural limits to need (1)	2

Question	Marking guidance	Marks
01.4	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> • Climate: warming increases evaporation, cloud and albedo; higher CO₂ increases photosynthesis and sequestration; hydrological cycle: more evaporation gives more precipitation; populations: density-dependent factors return numbers to carrying capacity (up to 2) • In agriculture soil degradation, water shortage and loss of predators would reduce yields (negative feedback limiting the activity) (up to 2) • Humans respond with fertilisers, irrigation and pesticides to maintain yields, further damaging the natural systems that support production (up to 2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	
02.1	<i>1 mark for each point.</i> • Negative feedback responds to any change by re-establishing the previous equilibrium (1) • Only large unusual events overwhelm the feedback so a new equilibrium forms (1) • Example: methane hydrate release at the end of the Permian causing climate change and mass extinction (1)	3
02.2	<i>1 mark for each point.</i> • In a diverse system each component is relatively less important so the loss / change of one has a small effect; alternative food sources and relationships buffer the change (1) • Natural: coral reefs and tropical rainforests with complex food webs are stable; simple polar / desert systems fluctuate (1) • Human: agroecosystems focus on a few crops and remove detritivores, pollinators and predators so are vulnerable to pests, disease, weather (1) • Diverse technical systems: a society using a range of renewable energy resources is more secure than one relying on a single source (1)	4
02.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Melting permafrost releasing methane (1) • Reduced albedo as ice / snow melt (1) • Methane hydrate release (1) • Forest and peat fires (1) • Faster soil decomposition (1) • Ocean acidification reducing sequestration (1) • Cirrus cloud formation (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	
03.1	<i>1 mark for each point.</i> • Moral belief: protecting the environment is the 'right' thing to do regardless of benefit to people (1) • Enlightened self-interest: protecting the environment is done because it causes fewer difficulties (for people) than not doing so (1)	2
03.2	<i>1 mark for each point.</i> • Total impact = population size x the individual (per capita) impact of each person (1) • Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact (1)	2

Question	Marking guidance	Marks
03.3	Any 1 from the points below - 1 mark each, maximum 1. • Slash and burn agriculture (1) • Fishing (1) • Discharge of sewage / biodegradable pollutants (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1.	
04.1	1 mark for each point. • Negative (1) • Positive (1) • Negative (1) • Positive (1)	4
04.2	1 mark for each point. • The response to a change acts to re-establish / return the system to the previous equilibrium (1) • So the change does not become self-sustaining unless a large unusual event overwhelms the negative feedback (1)	2
04.3	1 mark for each point. • eg the possible release of methane hydrate at the end of the Permian era (1) • Overwhelmed the negative feedback mechanisms, causing climate change and a mass extinction / a new equilibrium (1)	2
04.4	1 mark for each point. • Soil degradation, reduced water availability and loss of natural pest predators would reduce yields (negative feedback curtailing the activity) (1) • Instead of reducing the activity, farmers use fertilisers, irrigation and pesticides, which further damage the natural systems that support food production (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	
05.1	1 mark for each point. • The new opportunity is overwhelmingly attractive (economic benefit) (1) • Lack of knowledge / understanding of the impacts (1) • The impacts occur elsewhere or much later, so are not evident to those who benefit (1)	3
05.2	1 mark for each point. • Cultural belief that resources should be exploited to satisfy needs only (1) • Problems of over-exploitation become obvious quickly as local supplies dwindle, so behaviour changes (1)	2
05.3	1 mark for each point. • Resources are brought in from other areas (1) • If one area is depleted, exploitation moves to another area (eg timber, fish, metals, energy) without the consumers noticing (1)	2
05.4	1 mark for each point. • Easier access to energy has made many new activities possible (1) • The scale of activities has greatly increased (1) • Toxic wastes are produced that cannot be decontaminated by natural processes (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-371; AQA spec 3.6.1.	

Question	Marking guidance	Marks
06.1	1 mark. Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs (1)	1
06.2	1 mark for each point. - Negative feedback resists change / returns the system towards its previous state (1) - eg increased temperature increases cloud cover and albedo, or higher CO₂ increases photosynthesis (1) - Positive feedback increases the change / makes it self-sustaining (1) - eg melting permafrost releasing methane, declining albedo as ice melts, forest and peat fires (1)	4
06.3	1 mark. Many species with a complex food web so the loss of one has a small effect; agroecosystems have few species and lack detritivores, pollinators and pest predators that maintain stability (1)	1
06.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Principles of natural systems: dynamic equilibria and negative feedback; diverse systems resistant to change; low energy-density renewable energy (solar-driven hydrological, carbon and nitrogen cycles); low-temperature processes using enzymes; material cycles with wastes as raw materials; small number of abundant elements; biodegradable non-toxic wastes - Human systems: non-renewable energy, high temperatures (Haber process, incineration, manufacturing), linear resource use, dispersal of minerals and mixtures that are hard to separate, persistent toxins (PCBs, organochlorines, toxic metals in electronics) - Applications: renewable energy and heat pumps, bioleaching, recycling and design for re-use, industrial symbiosis, organic and integrated agriculture, retaining hedgerows and pollinators, circular economy - Human responses that override feedback (fertilisers, irrigation, pesticides) and positive feedback triggered by human activity (permafrost, methane hydrates, fires); tipping points - Barriers: population size and affluence, attitudes of urban populations to distant resources, economics, need for international cooperation, existing infrastructure - Justified conclusion on how far natural principles can be applied	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-374; AQA spec 3.6.1-3.6.3.	